

Computer Graphics Completion Plan

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Reference Screenshot



Required Features

Path Traced Image (Blender):

- Scene must be in a .blend file
- Suitable materials used to render the scene in Cycles Engine
- Final approximation of the reference image

Rasterised Image:

- Inputs, able to read vertices, edges, and faces from OBJs, and materials from data files
- Outputs, Visualisation of render written to an image file, png, tga, or ppm.
- Point drawing
- Line drawing
- Triangle rasterization
- Back face culling
- Perspective projection
- Perspective camera

Ray Traced Image:

- Inputs, able to read vertices, edges, and faces from OBJs, and materials from data files

- Outputs, Visualisation of render written to an image file, png, tga, or ppm.
- Create and trace rays through the scene
- Vector and colour functions
- Ray-triangle intersection
- Hit context management
- Shading based on surface normal
- Perspective camera

Possible Additional Features

Path Traced Image (Blender):

- Camera parameters and scene setup should closely match reference image
- Materials should be accurate to real world counterparts
- Decimated meshes to reduce rendering time

Rasterised Image:

- Back face culling (Z-Buffering)
- Different light types, directional, spotlight, point light
- Textures
- Surface shaders
- MipMapping
- Depth-of-field
- Bloom
- HDRI support

Ray Traced Image:

- Different light types, directional, spotlight, point light
- Textures
- Diffused materials
- Metallic materials
- Dielectric materials
- Antialiasing
- Depth-of-field
- HDRI support

Completion Plan

Week	Dates	Feature
1	27 th Jan-2 nd Feb	General: - Complete Lab One
2	3 rd Feb-9 th Feb	General: - Complete Lab Two
3	10 th Feb-16 th Feb	General: - Complete Lab Three
4	17 th Feb-23 rd Feb	General: - Complete Lab Four - Complete Completion Plan

		Path Traced Image: - Start Blender scene Rasterised Image: - Obj mesh inputs - Point drawing - Line drawing - Triangle Rasterization Ray Traced Image: - Obj mesh inputs
5	24 th Feb-2 nd Mar	General: - Complete Lab Five Path Traced Image: - Add models, set up camera Rasterised Image: - Scene setup, move meshes into position
6	3 rd Mar-9 th Mar	General: - Complete Lab Six Path Traced Image: - Complete & refine scene Rasterised Image: - Lighting setup
7	10 th Mar-16 th Mar	General: - Complete Lab Seven Rasterised Image: - Textures - Back face culling - Surface shading
8	17 th Mar-23 rd Mar	General: - Complete Lab Eight Rasterised Image: - MipMapping - Depth-of-field - Bloom
9	24 th Mar-30 th Mar	General: - Complete Lab Nine Rasterised Image: - HDRI Support Ray Traced Image: - Lighting - Textures - Ray Tracer From Started Code
10	31 st Mar-6 th Apr	General: - Complete Lab Ten Ray Traced Image: - Metallic materials - Dielectric materials - Diffused materials

11	7 th Apr-13 th Apr	General: - Complete Lab Eleven Ray Traced Image: - Antialiasing - Depth-of-field
12	14 th Apr-20 th Apr	General: - Complete Lab Twelve Ray Traced Image: - HDRI support
13	21 st Apr-27 th Apr	General: - Contingency Week
14	28 th Apr-4 th May	General: - Resolve Remaining Issues
N/A	4 th May-11 th May	General: - Submission Preparation & Hand-in
N/A	12 th May-16 th May	General: - Viva